CDA 106
Numerical Methods for Data Science

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Your classes on 13th Feb & 20th Feb will be held from 7:30AM — 9:30PM.

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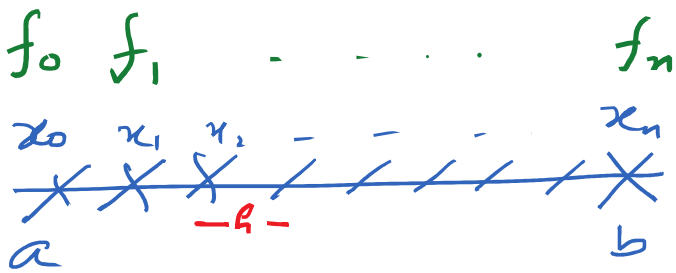
Other classes as per T.T.

Newton - Cotes formulae

$$\int_a^b f(x)\,dx$$

- Rectangular
- Trapezoidal
- Simpson's $\frac{1}{3}$rd
- Simpson's $\frac{3}{8}$th

$$\int_a^b f(x)\,dx = \sum_{i=0}^{n} \lambda_i f(x_i)$$



$$\frac{b-a}{n} = h$$

n equal parts

$n+1 \rightarrow$ points

$f_i = f(x_i)$

$$\int_a^b f(x)\,dx = \frac{h}{3}\left[f_0 + 4(f_1+f_3+\cdots) + 2(f_2+f_2+\cdots) + f_n\right]$$

$$= \underbrace{\frac{h}{3}}_{\lambda_0} f_0 + \underbrace{\frac{4h}{3}}_{\lambda_1} f_1 + \underbrace{\frac{2h}{3}}_{\lambda_2} f_2 + \cdots + \underbrace{\frac{h}{3}}_{\lambda_n} f_n$$

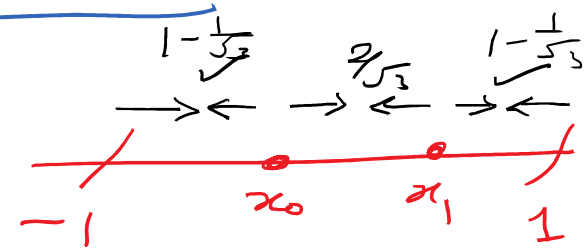
$$= \sum_{i=0}^{n} \lambda_i f_i$$

- "Gaussian Quadrature"
- Gaussian Integration formulae

$\Big\}$ $\parallel$

1. Gauss-Legendre Two-point rule (method)

$$\int_{-1}^{1} f(x)\,dx = \underset{=1}{\lambda_0} f(x_0) + \underset{=1}{\lambda_1} f(x_1)$$

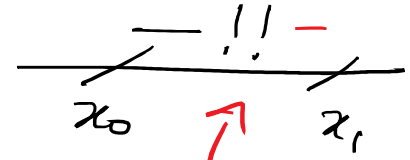


$x_0 = -\frac{1}{\sqrt{3}}$ $\quad x_1 = \frac{1}{\sqrt{3}}$

$$\boxed{\int_{-1}^{1} f(x)\,dx = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(1/\sqrt{3}\right)}$$

Details will be discussed later.

— x —



not same

Example:

$$\int_{-1}^{1} (x+1)\,dx = ?$$

Exact

$$\int_{-1}^{1} (x+1)\,dx = x\Big|_{-1}^{1} = \underline{\underline{2}}$$

Sol^n

$$f(x) = \underline{x+1}$$

$$f\left(-\frac{1}{\sqrt{3}}\right) = -\frac{1}{\sqrt{3}} + 1$$

$$f\left(1/\sqrt{3}\right) = \frac{1}{\sqrt{3}} + 1$$

Gauss-Legendre two point method.

$$\int_{-1}^{1} (x+1)\,dx = \left(-\frac{1}{\sqrt{3}} + 1\right) + \frac{1}{\sqrt{3}} + 1 = \underline{\underline{2}}.$$

Ex 2 $\quad I = \int_{-1}^{1} \frac{1}{x+3}\,dx$

$$I = \int_{-1}^{1} \frac{1}{x+3}\,dx = \ln(x+3)\Big|_{-1}^{1} = \ln 2 = 0.69$$

Gauss Legendre two point

$$f(x) = \frac{1}{x+3} \Rightarrow f\left(-\frac{1}{\sqrt{3}}\right) = \frac{1}{-\frac{1}{\sqrt{3}}+3}$$

$$\& \; f\left(\frac{1}{\sqrt{3}}\right) = \frac{1}{\frac{1}{\sqrt{3}}+3}$$

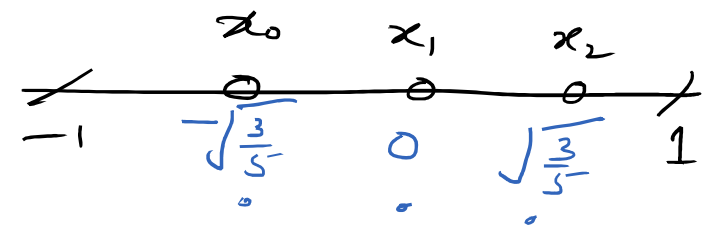
$$\int_{-1}^{1} \frac{1}{x+3}\,dx = \frac{1}{3-\frac{1}{\sqrt{3}}} + \frac{1}{3+\frac{1}{\sqrt{3}}} = \frac{3+\cancel{\frac{1}{\sqrt{3}}}+3-\cancel{\frac{1}{\sqrt{3}}}}{3^2-\frac{1}{3}} = \frac{18}{26}$$

$$= 0.69$$

Therefore, two point Gauss Legendre works pretty well !!

2) Gauss Legendre 3 point method

$$\int_{-1}^{1} f(x)\,dx = \lambda_0 f(x_0) + \lambda_1 f(x_1) + \lambda_2 f(x_2)$$



$\lambda_0 = \frac{5}{9}$, $\lambda_1 = \frac{8}{9}$ $\lambda_2 = \frac{5}{9}$

$x_0 = -\sqrt{3/5}$ $x_1 = 0$ $x_2 = \sqrt{3/5}$

$$\boxed{\int_{-1}^{1} f(x)\,dx = \frac{5}{9} f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\frac{3}{5}}\right)}$$

Ex 3 $\quad I = \int_{-1}^{1} (x+1)\,dx$

Gauss Legendre 3 point method.

$$f(x) = x + 1$$

$$\int_{-1}^{1} (x+1)\,dx = \frac{5}{9}\left[-\sqrt{\frac{3}{5}} + 1\right] + \frac{8}{9}\left[0+1\right] + \frac{5}{9}\left[\sqrt{\frac{3}{5}} + 1\right]$$

$$= \frac{5}{9} + \frac{8}{9} + \frac{5}{9} = \frac{18}{9} = 2$$

Can we apply above two methods to find

$$\int_a^b f(x)\,dx \;??$$

when $a \neq -1$ & $b \neq 1$

YES

$$\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$$

we know.

How??

$$I = \int_a^b f(t)\,dt$$

Transformation

$t \longrightarrow x$

$t = a \Rightarrow x = -1$

$t = b \Rightarrow x = +1$

$t = \alpha x + \beta$

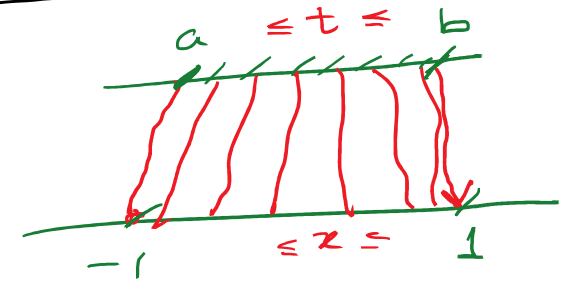
$a = -\alpha + \beta$

$b = \alpha + \beta$

$\alpha = \frac{b-a}{2}$

$\beta = \frac{a+b}{2}$

$$\boxed{t = \frac{b-a}{2}x + \frac{a+b}{2}}$$



Transformation

$t \longrightarrow x$

$$I = \int_{-1}^{1} f(x)\,dx$$

Gauss Legendre

•) Note carefully that the forms of "f(t)" & "f(x)" in the two integrals may be differe

Example 4

Find $I = \int_0^1 \frac{1}{x+1}\,dx$

$$I = \ln(x+1)\Big|_0^1 = \ln 2 = 0.69\ldots$$

Imp

$$I = \int_0^1 \underbrace{\frac{1}{t+1}}_{f(t)}\,dt \qquad a=0 \quad b=1$$

$$f(t) = \frac{1}{t+1}$$

As mentioned earlier, the forms of $f(t) = \frac{1}{t+1}$ & $f(x) = \frac{1}{x+3}$ are different.

$$t = \frac{1-0}{2}x + \frac{1+0}{2} = \frac{x+1}{2} \Rightarrow dt = \frac{dx}{2}$$

Therefore

$$I = \int_{-1}^{1} \frac{1}{\left(\frac{x+1}{2}\right)+1} \cdot \frac{dx}{2}$$

$$= \int_{-1}^{1} \underbrace{\left(\frac{1}{x+3}\right)}_{f(x)}\,dx$$

Gauss Legendre 2 point

$$= 0.69$$

(Refer to example 2 on slide 6)

Exercise Find $I = \int_0^2 \frac{dx}{x^2+2x+10}$ using two point Gauss Legendre formula!

$a = 0 \qquad b = 2$

Imp

$$I = \int_0^2 \frac{dt}{t^2+2t+10}$$

let $t = \frac{2-0}{2}x + \frac{2+0}{2} = x+1$

$dt = dx$

$$= \int_{-1}^{1} \frac{dx}{(x+1)^2+2(x+1)+10}$$

$$= \int_{-1}^{1} \frac{dx}{x^2+4x+13} \leftarrow$$ Gauss Legendre 2 point

$f(x) = \frac{1}{x^2+4x+13}$

$$= \frac{1}{\left(-\frac{1}{\sqrt{3}}\right)^2+4\left(-\frac{1}{\sqrt{3}}\right)+13} + \frac{1}{\left(\frac{1}{\sqrt{3}}\right)^2+4\left(\frac{1}{\sqrt{3}}\right)+13} = \underline{\underline{0.1546}}$$

Exercise Find $I = \int_0^1 \frac{1}{1+x^2}\,dx$

$I = \pi/4$
0.78539

1) Gauss Legendre 2 point

2) Gauss Legendre 3 point

$a =$ $b =$

$t = \frac{x+1}{2}$

Ans $I = \int_0^1 \frac{1}{1+t^2}\,dt = \int_{-1}^1 \frac{2\,dx}{x^2+2x+5}$

2 point 0.78688

3 point 0.78526

Exercise find $I = \int_1^2 \frac{2x}{1+x^4}\,dx$

using i) Gauss Legendre 2 point 0.5433 - -

ii) Gauss Legendre 3 point 0.5405

Ans: $a=1$ $b=2$ $t = \frac{x+3}{2}$ $dt = \frac{dx}{2}$

$$I = \int_{-1}^{1} \underbrace{\frac{8(x+3)}{16+(x+3)^4}}_{f(x)}\,dx \xrightarrow{\text{2 point}} = f\left(-\tfrac{1}{\sqrt{3}}\right) + f\left(\tfrac{1}{\sqrt{3}}\right)$$

$$= 0.5433$$

3 point

$$= \frac{5}{9} f\left(-\sqrt{\tfrac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\tfrac{3}{5}}\right) = 0.5405$$

$x^2 = u$

$$\int_1^4 \frac{du}{1+u^2}$$

$$= \tan^{-1} 4 - \tan^{-1} 1$$

$$= 0.540419$$

Exact Sol.